

EE 102 Week 1, Lecture 1 (Fall 2026)

Date: August 31, 2026

Learning Outcomes

By the end of this lecture, you should be able to:

- Understand the properties of signals
- Sketch signals in a 2D plane.

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1 Continuous-time and discrete-time signals

Continuous-time domain is $\mathbb{R}$, and we write continuous-time signals as $x(t)$, if they are continuous functions of time (recall: continuous functions from your math classes). On the other hand, discrete-time domain is $\mathbb{Z}$, and a discrete-time signal is written as $x[n]$, where $n \in \mathbb{Z}$. This means that discrete-time signals are defined only at integer time indices.

1.1 Sketching signals

To sketch, draw and label axes, mark key values (peaks, zeros, discontinuities), and indicate any symmetry, periodicity, decay/growth, or piecewise structure (try to identify as many properties as you can before starting to sketch). The best way to start your sketch is to compute the values of the signal at “easy” points like, zero, the max time, etc.

2 Properties of signals

2.1 Scaling

Time scaling changes the horizontal axis by a constant α :

$$x_s(t) = x(\alpha t).$$

If $0 < \alpha < 1$, the signal expands in time whereas if $\alpha > 1$, it compresses the signal in time.

2.2 Offset

Time shifting offsets the horizontal axis by a constant T . A (right) delay of T seconds is defined by

$$x_d(t) = x(t - T).$$

Equivalently, $x_d(t_1 + T) = x(t_1)$ for every t_1 .

Worksheet #2: Transforming Signals (Part 2) — Groups of 2.

Each pair of students should scale and shift a signal of their choice. Agree as a pair what scaling and shifting would mean and then draw it out. Clearly show the parameters and the transformed axes.

